

P-001: Unified HDC-LTC Neuron with Closed-Form Temporal Jumps

Provisional Patent Application

1. Title

Unified Hyperdimensional Computing Neuron with Liquid Time-Constant Dynamics and Closed-Form Temporal Jump Solutions Using Algebraic Binding Operations and SIMD-Accelerated Fused Evolution Kernels

2. Inventor(s)

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3. Date of Conception

2025 (estimated). First committed implementation: February 5, 2026. Conceptual design and architecture predate the initial commit.

4. Technical Field

This invention relates to artificial neural network architectures for temporal sequence processing, and more specifically to a neuron architecture that unifies hyperdimensional computing (HDC) with liquid time-constant (LTC) neural dynamics, enabling $O(1)$ closed-form temporal jumps using algebraic binding operations instead of matrix multiplication.

5. Abstract

A novel neural processing unit is disclosed that represents its internal state as a high-dimensional continuous hypervector (e.g., 16,384 dimensions) and evolves that state through liquid time-constant dynamics using hyperdimensional algebraic operations—specifically element-wise binding and normalized bundling—in place of conventional weight matrix multiplications. This substitution reduces the per-neuron parameter count from $O(D^2)$ to $O(D)$ while preserving the expressive power needed for temporal sequence modeling. The neuron’s governing ordinary differential equation admits a closed-form exponential decay solution, enabling $O(1)$ temporal jumps to arbitrary future time horizons at cost independent of the time step magnitude. A state-dependent time constant adapts the neuron’s response speed to the complexity of its current representation. An adaptive gating mechanism, inspired by Closed-form Continuous-depth (CfC) networks, computes a learned interpolation factor that blends the current state with the computed equilibrium. Optional Fourier basis injection provides time-varying perturbations for richer temporal dynamics. Hand-written AVX2+FMA SIMD kernels fuse the entire equilibrium computation and state interpolation into a

single pass through the dimension, eliminating intermediate allocations (256 KB per invocation for $D=16,384$) and processing 8 elements per cycle. The architecture supports multiple learning rules including Hebbian, contrastive, STDP, triplet, and full backpropagation through time (BPTT) with analytical gradient computation through the closed-form step.

6. Background and Prior Art

6.1 Liquid Time-Constant (LTC) Networks Lechner et al. (2020, “Neural Circuit Policies Enabling Auditable Autonomy,” Nature Machine Intelligence) introduced LTC neurons whose time constants are input-dependent, governed by the ODE:

$$dx/dt = (-x + f(Wx + Uu)) / \tau$$

where W and U are weight matrices, f is a nonlinear activation, and τ is a time constant. These networks demonstrated strong performance on temporal tasks but require numerical ODE integration (Euler or RK4), making them computationally expensive for large time horizons.

6.2 Closed-form Continuous-depth (CfC) Networks Hasani et al. (2021, “Liquid Time-constant Networks,” AAAI) and subsequent work on CfC networks derived closed-form solutions for LTC dynamics, enabling single-step temporal updates. However, CfC networks retain conventional weight matrices ($O(D^2)$ parameters per layer) and do not exploit the algebraic structure of hyperdimensional computing.

6.3 Hyperdimensional Computing (HDC) Kanerva (2009, “Hyperdimensional Computing: An Introduction to Computing in Distributed Representation with High-Dimensional Random Vectors”) established the theoretical foundation for computing with high-dimensional vectors using binding (element-wise multiplication) and bundling (normalized addition). Rahimi et al. (2017) and subsequent work applied HDC to classification tasks. However, existing HDC systems are typically static classifiers without temporal dynamics—they lack the continuous-time evolution needed for sequence processing.

6.4 Standard Recurrent Networks (LSTM/GRU) Long Short-Term Memory (Hochreiter and Schmidhuber, 1997) and Gated Recurrent Units (Cho et al., 2014) are the dominant architectures for temporal processing. Both use weight matrices with $O(D^2)$ parameters and cannot perform temporal jumps—they must process every time step sequentially.

6.5 Gap in Prior Art No prior art combines HDC algebraic operations with LTC temporal dynamics. Specifically:

- **LTC/CfC networks** use matrix multiplication ($O(D^2)$ parameters), not HDC binding ($O(D)$ parameters).
- **HDC systems** are static classifiers without continuous-time ODE dynamics.
- **No existing architecture** provides both $O(D)$ parameterization AND $O(1)$ temporal jumps.
- **No existing architecture** uses state-dependent time constants computed from hypervector norms.
- **No existing SIMD implementation** fuses HDC binding, bundling, activation, and temporal interpolation into a single pass.

7. Detailed Technical Description

7.1 Neuron Architecture The unified HDC-LTC neuron (`HdcLtcUnifiedNeuron`) maintains the following internal hypervectors, all of dimension D (default $D = 16,384$):

Component	Symbol	Role
State	$\mathbf{x}$	Current neuron state (<code>ContinuousHV</code>)
Weight HV	$\mathbf{W}$	State transformation via binding (replaces weight matrix)
Input Mask	$\mathbf{U}$	Input transformation via binding
Tau Modulator	$\mathbf{T_mod}$	Input-dependent time constant adjustment
Gate Weight	$\mathbf{G_w}$	Gating function for closed-form interpolation
Gate Bias	$\mathbf{G_b}$	Gating bias (initialized at 0.1x scale)
Weight Momentum	$\mathbf{M_w}$	Momentum accumulator for weight updates
Input Momentum	$\mathbf{M_u}$	Momentum accumulator for input mask updates

All internal HVs are initialized via Gram-Schmidt orthogonalization (`ContinuousHV::orthogonal_set`) to ensure pairwise similarity < 0.02 at initialization, minimizing interference between learned representations.

Key insight: By replacing weight *matrices* ($D \times D$) with weight *hypervectors* (D), each neuron requires $O(D)$ parameters instead of $O(D^2)$. For $D = 16,384$, this is a reduction from ~268 million to ~16,384 parameters per weight component.

7.2 Governing ODE The neuron's dynamics are governed by:

$$dx/dt = (-x + f(W * x + U * u)) / \tau(||x||)$$

In HDC notation with explicit operations:

$$dx/dt = (-x \text{ (bundling-inverse) } f(W \text{ (bind) } x \text{ (bundle) } U \text{ (bind) } u)) / \tau(||x||)$$

Where: - (bind) denotes HDC binding: element-wise multiplication, $(A \text{ bind } B)_i = A_i * B_i$ - (bundle) denotes HDC bundling: normalized element-wise addition, $\text{bundle}(A, B) = (A + B) / n$ where n is the number of operands - f is a configurable activation function (`Tanh`, `Sigmoid`, `SiLU`, `Identity`, or `BoundedTanh`)

7.3 Equilibrium State Computation The equilibrium state x_{inf} is computed as:

```
x_inf = f(bundle(W (bind) x, U (bind) u [, F(t)]))
```

Where $F(t)$ is an optional Fourier basis injection (Section 7.7). The `compute_equilibrium` method performs:

1. **State transformation:** `transformed_state = W (bind) x` (element-wise multiply)
2. **Input transformation:** `masked_input = U (bind) u` (element-wise multiply)
3. **Bundling:** `combined = bundle(transformed_state, masked_input)` (normalized sum)
4. **Activation:** `x_inf = f(combined)` (element-wise nonlinearity)

7.4 State-Dependent Time Constant The time constant adapts to both state complexity and input characteristics:

```
tau(x, u) = tau_0 * (1 + backbone * ||x||) * (1 + 0.2 * sim(u, T_mod))
```

Where: - `tau_0` is the base time constant (default: 0.1s, i.e., 100ms) - `backbone` is the scaling factor (default: 0.5) - `||x||` is the L2 norm of the state hypervector - `sim(u, T_mod)` is the cosine similarity between input and tau modulator HV - The result is clamped to [0.01, 10.0] seconds

This means the neuron responds faster when its state is simple (low norm) and slower when processing complex representations (high norm), providing an adaptive computational budget.

7.5 Closed-Form Solution For the ODE $dx/dt = (x_{\text{inf}} - x) / \tau$, assuming x_{inf} is approximately constant over the time step, the exact analytical solution is:

```
x(t + dt) = x_inf + (x(t) - x_inf) * exp(-dt / tau)
```

This can be rewritten as an interpolation:

```
x(t + dt) = sigma * x_inf + (1 - sigma) * x(t)
```

where `sigma = 1 - exp(-dt / tau)`.

Critical property: The computational cost is $O(D)$ regardless of dt . A jump of 0.001 seconds costs exactly the same as a jump of 100 seconds.

7.6 Adaptive Gating Mechanism (CfC-style) The `compute_gating` method computes a learned interpolation factor `sigma` that incorporates both the exponential decay and a learned gating signal:

1. **Fused bundle + similarity:** Compute `bundle(x, u)` and its similarity with `G_w` in a single inline loop (avoiding a 16,384-element allocation):

```
gate_activation = sim(bundle(x, u), G_w) + mean(G_b)
```

2. **Sigmoid gating with steepness control:**

```
sigma_base = 1 / (1 + exp(-gate_activation * steepness))
```

where `steepness` (default: 1.0) controls the sharpness of the gating decision.

3. **Time-scaled gating:** Combine learned gating with exponential decay:

```
decay = exp(-dt / tau)
sigma = 1 - decay * (1 - sigma_base)
```

Result is clamped to [0.0, 1.0].

4. **Final interpolation** (zero-allocation in-place):

```
x(t + dt) = (1 - sigma) * x(t) + sigma * x_inf
```

Implemented via `lerp_in_place` which modifies state values directly without allocating a new hypervector.

7.7 Fourier Basis Injection When configured with one or more frequencies, the neuron computes a time-varying Fourier basis hypervector that is bundled into the equilibrium computation as a third component:

```
x_inf = f(bundle(W (bind) x, U (bind) u, F(t)))
```

The Fourier basis $F(t)$ is constructed by distributing sin/cos pairs across the dimension via a strided pattern:

- For each frequency f_i , compute $\sin(2\pi f_i t) * \text{amplitude}$ and $\cos(2\pi f_i t) * \text{amplitude}$
- Sin values are placed at dimension indices $2*i$, $2*i + \text{total_channels}$, $2*i + 2*\text{total_channels}$, ...
- Cos values are placed at dimension indices $2*i+1$, $2*i+1 + \text{total_channels}$, ...
- $\text{total_channels} = 2 * \text{num_frequencies}$
- Default amplitude: 0.1 (small perturbation, not a driver)

This enables the neuron to encode absolute time information and exhibit periodic behaviors without external clock signals.

7.8 SIMD-Optimized Fused Evolution Kernel The `evolve_closed_form_fused` method eliminates all intermediate allocations by fusing the entire evolution step into a single pass through the dimension. On x86_64 with AVX2+FMA, hand-written SIMD intrinsics process 8 f32 elements per cycle (256-bit registers).

Fused Tanh kernel (`fused_tanh_avx2`):

For each 8-element chunk: 1. Load `state[i]`, `weight[i]`, `input_mask[i]`, `input[i]` into AVX2 registers 2. Compute pre-activation: `pre_act = (W[i] * state[i] + M[i] * input[i]) * pre_scale` using FMA instructions (`_mm256_fmadd_ps`) 3. Compute `fast_tanh` via rational approximation: `tanh(x) approx x * (27 + x^2) / (27 + 9*x^2)` for $|x| < 4.97$, with ± 1.0 clamping for $|x| \geq 4.97$ (max error ~ 0.004) 4. Lerp: `new_state[i] = (1-sigma) * state[i] + sigma * x_inf[i]` using FMA 5. Store result back to state

A scalar remainder loop handles `dim % 8` trailing elements.

Memory savings: For $D = 16,384$, the fused kernel eliminates 4 intermediate ContinuousHV allocations per call, saving $4 * 16,384 * 4 \text{ bytes} = \mathbf{256 \text{ KB per invocation}}$.

A separate `fused_identity_avx2` kernel handles the Identity activation case (no tanh computation).

Fast tanh approximation (`fast_tanh`):

```
fast_tanh(x) = x * (27 + x^2) / (27 + 9*x^2)    if |x| < 4.97
              = signum(x)                      if |x| >= 4.97
```

This rational approximation is pure arithmetic (no branches in the hot path, no libm calls), enabling LLVM to auto-vectorize the scalar fallback path and directly mapping to SIMD arithmetic in the hand-written kernels. Maximum error is ~ 0.004 (0.4%).

7.9 Multiple Evolution Methods The architecture provides five evolution methods with different accuracy/performance tradeoffs:

Method	Complexity	Accuracy	Allocations
evolve (Euler)	$O(D)$ per step	First-order	2 HVs
evolve_rk4	$O(4D)$ per step	Fourth-order	8 HVs
evolve_closed_form(D), dt-independent	$O(D)$	Gated approximation	1 HV (equilibrium)
evolve_closed_form(D), dt-independent	$O(D)$	Gated approx ($\sim 0.4\%$ tolerance)	0 HVs
evolve_closed_form(D), dt-independent	$O(D)$	Exact for linear ODE	2 HVs
evolve_closed_form(Iterative)	$O(D)$	High (sub-stepped)	2 HVs per sub-step

7.10 Learning and Adaptation The neuron supports six learning rules, all operating on hypervector weights via HDC operations:

1. **Hebbian update:** $\text{delta_W} = \text{lr} * (\text{input}(\text{bind}) \text{ state})$, with momentum (default $\text{beta}=0.9$) and L2 weight decay (default 0.0001). Weight norm clipped to 2.0.
2. **Contrastive update:** Pulls state toward positive examples and pushes away from negative examples. Gradient computed as $\text{W}(\text{bind})(\text{positive} - \text{state}) + 0.5 * \text{W}(\text{bind})(\text{state} - \text{negative})$.
3. **STDP (Spike-Timing Dependent Plasticity):** Asymmetric exponential windows with $\text{tau_plus} = \text{tau_minus} = 20\text{ms}$, $\text{A_plus} = 1.0$, $\text{A_minus} = 0.5$. Pre-before-post strengthens, post-before-pre weakens.
4. **Adaptive (Adam-like):** First-moment estimation with bias correction. Uses momentum beta1 (default 0.9) with correction factor $1 / (1 - \text{beta1}^t)$.
5. **Regularized Hebbian:** Combines Hebbian learning with homeostatic plasticity (target activity scaling, clamped [0.5, 2.0]) and explicit L2 penalty term.
6. **Triplet loss:** Metric learning with $\text{loss} = \max(0, \text{dist_pos} - \text{dist_neg} + \text{margin})$. Updates weights only when the margin is violated.
7. **BPTT (Backpropagation Through Time):** The backward method computes analytical gradients through the closed-form step:
 - Forward recomputation of $z, x_{\text{inf}}, \text{tau}, \text{sigma}, \text{new_state}$
 - MSE loss gradient: $dL/dx' = 2(x' - \text{target}) / D$
 - Through interpolation: $dL/dx_{\text{inf}} = dL/dx' * \text{sigma}$
 - Through activation: $dL/dz = dL/dx_{\text{inf}} * f'(z)$ (element-wise)
 - Weight gradient via binding chain rule: $dL/dW = dz(\text{bind}) x$ (since binding is element-wise, the derivative is the other operand)

- Input mask gradient: $dL/dU = dz \text{ (bind) } u$
- Tau gradient (scalar): $dL/d\tau = \sum_i (dh_i * (x_{inf_i} - x_i)) * (-dt/\tau^2) * \exp(-dt/\tau)$
- Input gradient for inter-layer BPTT: $dL/du = dz \text{ (bind) } U$

The `apply_gradients` method performs SGD with momentum, weight decay, and norm clipping to 2.0.

7.11 Network Architecture The `HdcLtcUnifiedNetwork` composes multiple neurons into a layered architecture:

- Configurable layer sizes (e.g., [4, 8, 4])
- Inter-layer binding vectors for representational transformation between layers
- Optional skip connections from input to deeper layers
- Layer output computed as mean of neuron states within each layer
- Supports both standard evolution and closed-form evolution across the full network
- Deterministic initialization from a genesis seed for reproducibility

7.12 Numerical Stability

- **State bounding:** State norm is soft-clipped to 5.0 after every evolution step
- **Exponent clamping:** The decay exponent $(-dt/\tau)$ is clamped to a minimum of -87.0 to prevent f32 underflow (since $\exp(-87) \approx 1.6e-38 > 0$)
- **Weight norm clipping:** All weight HVs are clipped to norm 2.0 after every learning update
- **Running statistics:** Exponential moving average ($\alpha=0.01$) of state norm for monitoring

8. Novelty Statement

The following aspects are believed to be new relative to all known prior art:

- **Hypervector state with LTC dynamics:** No prior system uses a high-dimensional hypervector as the state of a liquid time-constant neuron. Prior LTC/CfC neurons use scalar or low-dimensional vector states with weight matrices.
- **HDC binding replacing weight matrices:** The use of element-wise binding (multiplication) between weight hypervectors and state/input hypervectors to replace matrix multiplication reduces parameters from $O(D^2)$ to $O(D)$ per weight component while preserving the near-orthogonality properties of HDC.
- **Closed-form solution for HDC-state ODEs:** The derivation and implementation of an exact analytical solution $x(t+dt) = x_{inf} + (x(t) - x_{inf}) * \exp(-dt/\tau)$ where x, x_{inf} are hypervectors of dimension D , enabling $O(1)$ temporal jumps independent of dt .
- **State-dependent time constant from hypervector norm:** The formula $\tau(x) = \tau_0 * (1 + backbone * ||x||) * (1 + 0.2 * \text{sim}(u, T_{mod}))$ where $||x||$ is the L2 norm of the D -dimensional state hypervector, providing adaptive computational pacing based on representational complexity.
- **Fused SIMD evolution kernel:** A single-pass AVX2+FMA kernel that computes binding, bundling, rational-approximation tanh, and temporal interpolation without any intermediate

memory allocations, saving 256 KB per invocation for $D=16,384$.

- **Rational fast_tanh approximation for SIMD:** The specific formula $\tanh(x) \approx x * (27 + x^2) / (27 + 9*x^2)$ chosen for its pure-arithmetic nature enabling auto-vectorization and direct SIMD mapping.
- **Fourier basis injection into HDC equilibrium:** Bundling time-varying sinusoidal signals (distributed across dimensions via strided placement) as an additional component in the equilibrium computation, providing absolute time encoding without external clock inputs.
- **BPTT through closed-form HDC step:** Analytical gradient computation exploiting the fact that HDC binding is element-wise multiplication, yielding the simple chain rule $d(A \text{ bind } B)/dA = B$ element-wise.
- **Gram-Schmidt orthogonal initialization of internal HVs:** Initializing all 5 internal hypervectors (W, U, T_mod, G_w, G_b) via modified Gram-Schmidt to ensure pairwise similarity < 0.02 , minimizing initialization interference.

9. Suggested Claims

Claim 1 (independent): A method for neural processing comprising: (a) maintaining a neuron state as a continuous-valued hypervector of dimension D ; (b) computing an equilibrium hypervector by applying hyperdimensional binding operations between weight hypervectors and the state hypervector, bundling the result with a bound input hypervector, and applying a nonlinear activation function; (c) computing a state-dependent time constant based on the L2 norm of the state hypervector; and (d) updating the state hypervector using a closed-form exponential decay solution that interpolates between the current state and the equilibrium in $O(D)$ operations independent of the time step magnitude.

Claim 2 (independent): A neural processing unit comprising: a state register storing a continuous-valued hypervector of dimension D ; a weight hypervector of dimension D ; an input mask hypervector of dimension D ; and processing logic configured to evolve the state hypervector by computing element-wise binding between the weight hypervector and the state hypervector, computing element-wise binding between the input mask hypervector and an input hypervector, computing a normalized sum of the binding results, applying an activation function, and interpolating between the current state and the activated result using an exponential decay factor dependent on a state-derived time constant.

Claim 3 (independent): A computer-implemented method for temporal sequence processing comprising: representing each neuron state as a D -dimensional hypervector where D is at least 1,000; replacing weight matrix multiplications with element-wise hyperdimensional binding operations between weight hypervectors and operand hypervectors; computing temporal state updates using the closed-form solution $x(t+dt) = x_inf + (x(t) - x_inf) * \exp(-dt / \tau(||x||))$; and performing said computation in a single SIMD-fused pass through the dimension that eliminates all intermediate hypervector allocations.

Claim 4 (dependent on 1): The method of claim 1, wherein the state-dependent time constant is computed as: $\tau(x, u) = \tau_0 * (1 + backbone * ||x||) * (1 + \alpha * \sin(u, T_mod))$, where τ_0 is a base time constant, $backbone$ is a scaling factor, $||x||$ is the L2 norm of the state

hypervector, sim denotes cosine similarity, u is the input hypervector, T_mod is a learned tau modulator hypervector, and α is a constant (e.g., 0.2).

Claim 5 (dependent on 1): The method of claim 1, further comprising computing an adaptive gating factor σ by: bundling the state and input hypervectors; computing cosine similarity between the bundle and a gate weight hypervector; applying a sigmoid function with configurable steepness; and combining the sigmoid output with an exponential decay factor as $\sigma = 1 - \exp(-dt/\tau) * (1 - \sigma_base)$.

Claim 6 (dependent on 3): The method of claim 3, wherein the SIMD-fused pass uses AVX2 256-bit registers with FMA instructions to: load 8 elements of state, weight, $input_mask$, and input per cycle; compute binding and bundling via fused multiply-add; compute a rational tanh approximation as $x(27+x^2)/(27+9x^2)$; and compute the temporal interpolation via fused multiply-add, all without writing any intermediate hypervectors to memory.

Claim 7 (dependent on 1): The method of claim 1, further comprising injecting a Fourier basis hypervector into the equilibrium computation by: for each of K configured frequencies f_k , computing $\sin(2\pi f_k t)$ and $\cos(2\pi f_k t)$ scaled by an amplitude parameter; distributing the sin and cos values across the D dimensions using a strided pattern with stride $2K$; and bundling the Fourier basis hypervector as an additional component in the equilibrium computation.

Claim 8 (dependent on 1): The method of claim 1, further comprising performing backpropagation through the closed-form temporal step by: computing a loss gradient with respect to the updated state; propagating the gradient through the interpolation to obtain a gradient on the equilibrium; propagating through the activation function using the element-wise derivative; and computing weight gradients using the property that the derivative of element-wise binding $A*B$ with respect to A is B .

Claim 9 (dependent on 2): The neural processing unit of claim 2, wherein the weight hypervector, input mask hypervector, a tau modulator hypervector, a gate weight hypervector, and a gate bias hypervector are initialized via Gram-Schmidt orthogonalization to have pairwise cosine similarity below 0.02.

Claim 10 (dependent on 1): The method of claim 1, further comprising applying a Hebbian learning rule by: computing a correlation hypervector as the element-wise binding of the input hypervector and the state hypervector; updating a momentum accumulator as $M = \beta * M + \text{lr} * \text{correlation}$; and updating the weight hypervector as $W = (1 - \text{decay}) * W + M$, with norm clipping to a maximum of 2.0.

Claim 11 (dependent on 1): The method of claim 1, further comprising applying a spike-timing dependent plasticity (STDP) rule by: computing a timing-dependent weight change Δw using exponential windows with time constants τ_plus and τ_minus ; computing a correlation hypervector as the element-wise binding of pre-synaptic and post-synaptic hypervectors; and updating the weight hypervector by the product of Δw and the correlation, modulated by a learning rate.

Claim 12 (dependent on 1): The method of claim 1, wherein the dimension D is at least 8,192 and the hyperdimensional binding operation is element-wise multiplication of f32 values, such that the parameter count per weight component is $O(D)$ rather than $O(D^2)$.

Claim 13 (dependent on 3): The method of claim 3, wherein the rational tanh approximation computes $\tanh(x)$ as $x(27+x^2)/(27+9x^2)$ for $|x| < 4.97$ and as $\text{signum}(x)$ for $|x| \geq 4.97$, achieving

maximum error of approximately 0.004 while enabling SIMD vectorization.

Claim 14 (dependent on 1): The method of claim 1, further comprising organizing multiple neuron instances into a layered network with inter-layer binding vectors, wherein the output of each layer is computed as the mean of its constituent neuron states, and inter-layer communication is mediated by binding the previous layer’s output with a layer-specific binding hypervector.

Claim 15 (dependent on 1): The method of claim 1, wherein the neuron supports multiple evolution modes including: Euler integration; fourth-order Runge-Kutta integration; single-step gated closed-form evolution; zero-allocation fused closed-form evolution; pure analytical exponential decay; and iterative closed-form sub-stepping with sub-step size of $\tau/10$.

10. Experimental Validation

All performance figures are from the implemented system running the Symthaea cognitive pipeline.

10.1 Cycle Performance

- **Full cognitive loop cycle time:** 4.3ms in release mode (234 Hz), exceeding the 50 Hz real-time target by 4.7x
- **Cycle with hypervector input (non-text):** 2.0ms/cycle (500 Hz)
- **Warm word encoding:** 97 ns
- **Sentence encoding:** 379 us for 10 words (only 12% of cycle time)

10.2 Memory Efficiency

- **Parameter count per weight component:** $O(D) = 16,384$ f32 values = 64 KB
- **Equivalent matrix parameterization:** $O(D^2) = 268,435,456$ f32 values = 1 GB
- **Compression ratio:** 16,384x fewer parameters per weight component
- **SIMD fused kernel savings:** 256 KB eliminated per evolution invocation (4 intermediate HV allocations removed)

10.3 Test Coverage

- **Tests in `hdc_ltc_unified.rs`:** 29 tests, all passing
- **Total project tests:** 7,586+ passing (3,961 symthaea main + 3,625 core)
- **Specific validations:**
 - Closed-form vs Euler convergence: similarity > 0.5 after 1s evolution
 - $O(1)$ property: 0.1s and 100.0s jumps both produce bounded, valid states
 - State-dependent tau: measurable tau change after 50 evolution steps
 - HDC binding dissimilarity: binding output < 0.3 similarity to either operand
 - Gating monotonicity: larger dt produces larger sigma
 - Orthogonal initialization: pairwise similarity < 0.02 for all 5 internal HVs
 - Numerical stability: no NaN/Inf after extreme $dt=100.0$ with $\tau=0.01$
 - Fourier injection: measurably different trajectories with vs without Fourier basis
 - Genesis determinism: bit-identical reconstructions from same seed

10.4 Broader System Validation

- **Moral algebra accuracy:** 91.1% on ethical dilemma classification
- **LibriSpeech benchmark:** 94.5% accuracy
- **ISOLET benchmark:** 91.7% accuracy
- **Psych-bench qualia confidence:** 0.683 composite (MODERATE), 7/7 predictions met
- **Voice synthesis:** Mean formant error 4.4 Hz (LTC-controlled), MCD 0.02 dB

11. Key Source Files

File	Description	LOC
symthaea-core/src/hdc/hdc_neurons.py	HDC neuron implementation	~2,403
symthaea-core/src/hdc/ConfinedHVC.py	ConfinedHVC type and HDC operations	–
symthaea-core/src/hdc/HDC.py	HDC module root, dimension constants (16,384)	–
symthaea/src/cognitive_loop/cognitive_pipeline.py	Cognitive pipeline using HDC-LTC neurons	–
symthaea-core/src/gensim/seed.py	Gensim deterministic initialization	–

12. Closest Prior Art References

1. Hasani, R., Lechner, M., Amini, A., et al. (2021). “Liquid Time-constant Networks.” *AAAI Conference on Artificial Intelligence*. – Introduces LTC neurons with input-dependent time constants and closed-form CfC solutions, but uses weight matrices ($O(D^2)$), not HDC binding.
2. Lechner, M., Hasani, R., et al. (2020). “Neural Circuit Policies Enabling Auditable Autonomy.” *Nature Machine Intelligence*, 2(10), 642-652. – Original LTC neuron formulation with ODE-based evolution; no closed-form solution, no HDC operations.
3. Kanerva, P. (2009). “Hyperdimensional Computing: An Introduction to Computing in Distributed Representation with High-Dimensional Random Vectors.” *Cognitive Computation*, 1(2), 139-159. – Foundational HDC framework with binding/bundling operations, but no temporal dynamics or ODE evolution.
4. Rahimi, A., Kanerva, P., Rabaey, J.M. (2017). “A Robust and Energy-Efficient Classifier Using Brain-Inspired Hyperdimensional Computing.” *ISLPED*. – HDC classification; static, no temporal dynamics.
5. Hochreiter, S. and Schmidhuber, J. (1997). “Long Short-Term Memory.” *Neural Computation*, 9(8), 1735-1780. – LSTM architecture with $O(D^2)$ parameters, no closed-form temporal jumps, no HDC operations.
6. Cho, K., et al. (2014). “Learning Phrase Representations using RNN Encoder-Decoder for Statistical Machine Translation.” *EMNLP*. – GRU architecture; same limitations as LSTM relative to this invention.
7. Ge, L. and Parhi, K.K. (2020). “Classification using Hyperdimensional Computing: A Review.” *IEEE Circuits and Systems Magazine*. – Survey of HDC classification methods; no temporal/ODE components.

8. Thomas, A., et al. (2021). “Theoretical Foundations of Hyperdimensional Computing.” *Journal of Artificial Intelligence Research*. – Theoretical analysis of HDC capacity and operations; no connection to continuous-time neural dynamics.
9. US Patent 11,537,868 (Hasani et al.) – “Liquid time-constant networks” – Covers LTC/CfC with conventional weight matrices; does not disclose HDC binding as a replacement for matrix multiplication.

13. Figures (Text Descriptions)

Figure 1: Unified HDC-LTC Neuron Architecture Diagram

A block diagram showing the internal architecture of a single `HdcLtcUnifiedNeuron`. At center, a large box labeled “State x (16,384D ContinuousHV)”. Inputs flow from the left: “Input u (16,384D)” passes through a binding operation (circle with $\times$ symbol) with “Input Mask U (16,384D)” to produce “Masked Input”. The state passes through a binding operation with “Weight HV W (16,384D)” to produce “Transformed State”. Both results enter a bundling operation (circle with $+$ symbol) labeled “Normalized Sum”, followed by an activation function block $f()$. The output is labeled “ x_{∞} (equilibrium)”. A separate path shows “State Norm $\|x\|$ ” feeding into a “Tau Computation” block that also receives similarity from the input and Tau Modulator. The tau value feeds into an “Exponential Decay / Gating” block that also receives the gating signal from Gate Weight and Gate Bias operating on a bundle of state and input. The final output arrow shows “ $\sigma * x_{\infty} + (1-\sigma) * x$ ” feeding back to the state. An optional dashed path shows “Fourier Basis $F(t)$ ” entering the bundling operation as a third component.

Figure 2: Closed-Form Temporal Jump Illustration

A time-axis diagram comparing three evolution strategies. The x-axis represents time from $t=0$ to $t=T$. Three rows: - **Top row (Euler)**: Many small arrows stepping from $x(0)$ toward x_{∞} , each requiring one $O(D)$ computation. Total: $N \text{ steps} * O(D)$. - **Middle row (RK4)**: Fewer but larger arrows, each with 4 sub-evaluations shown as dotted lines. Total: $N/4 \text{ steps} * O(4D)$. - **Bottom row (Closed-Form)**: A single curved arrow from $x(0)$ directly to $x(T)$, labeled “ $O(D)$, ONE STEP”. The exponential decay curve $x_{\infty} + (x(0) - x_{\infty}) * \exp(-T/\tau)$ is shown as a dashed line. A callout box emphasizes: “Same cost for $dt=0.001s$ and $dt=100s$.”

Figure 3: Parameter Efficiency Comparison Chart

A bar chart comparing parameter counts for a single neuron layer of dimension $D=16,384$: - **LSTM**: $4 * D^2 + 4 * D = \sim 1.07$ billion parameters (tall bar, log scale) - **GRU**: $3 * D^2 + 3 * D = \sim 805$ million parameters - **LTC/CfC**: $D^2 + D = \sim 268$ million parameters - **HDC-LTC Unified (this invention)**: $5 * D = \sim 82,000$ parameters (dramatically shorter bar) A secondary annotation shows the ratio: “16,384x fewer parameters than LTC/CfC per weight component.”

Figure 4: SIMD Fused Kernel Data Flow

A pipeline diagram showing the data flow through the `fused_tanh_avx2` kernel for one 8-element chunk. Eight parallel lanes are shown (one per f32 element in a 256-bit AVX2 register). The pipeline stages are: 1. **LOAD**: Four `_mm256_loadu_ps` operations load `state[i]`, `weight[i]`, `input_mask[i]`, `input[i]` 2. **BIND+BUNDLE**: `_mm256_mul_ps(w, s)` computes Wx , then `_mm256_fmadd_ps(m, inp, ws)` fuses $Uu + W*x$, then `_mm256_mul_ps(..., scale)` applies the 0.5 bundling normal-

ization 3. **FAST_TANH**: Three arithmetic operations: x^2 , $x*(27+x^2)$, $/(27+9*x^2)$, followed by `_mm256_blendv_ps` for the $|x|>4.97$ clamping 4. **LERP**: `_mm256_fmadd_ps(sigma, x_inf, _mm256_mul_ps(oms, s))` computes the final interpolation 5. **STORE**: `_mm256_storeu_ps` writes result back to state

A callout notes: “Zero intermediate allocations. 256 KB saved per call for D=16,384.”
